

Study Questions: Perception

1. Define **Perception** in your own words, specifically as it applies to Middle School.
2. How does the Free Energy Principle constrain our understanding of Perception?
3. Contrast the Classical view of Perception with the Active Inference view.
4. Explain how the "jacket in the dark" example shows that your brain predicts what you see before you actually see it.
5. What is a **prediction error**? Give an example of one you experienced this week -- big or small.
6. Why can you read scrambled words like "tihs snetnece"? What does this tell you about how your brain processes information?
7. What is the **spotlight effect**? Describe a time you felt like everyone was watching you. Was that feeling accurate?
8. Your friend sends you a text and you immediately think they are mad at you. Explain how your brain's predictions (not the actual text) could be causing that reaction.
9. If perception is based on predictions, can two people look at the same situation and honestly see different things? Give an example from school.
10. A student studies hard for a test and expects easy questions, but the test is much harder than expected. Is this a perception issue, a prediction error, or both? Explain.
11. How is your brain like a weather forecaster? What happens when the forecast is wrong?
12. Describe one perception trap (other than the spotlight effect) that middle schoolers commonly fall into. Why does it happen?
13. If your brain shows you its predictions instead of raw reality, how can you ever know what is "really" happening? What strategies help?

14. You expect your favorite song to play next on a playlist, but a different song starts.
Describe what happens in your brain during that moment of surprise.
15. Why does your first impression of someone new often say more about YOUR brain than about THEM?
16. A coach tells one group of players "this drill is really hard" and another group "this drill is pretty easy." Both groups do the same drill. Predict how each group will feel and explain why using perception concepts.
17. Create a step-by-step "Perception Check" process that someone could use when they have a strong emotional reaction to a situation.
18. How do optical illusions prove that your brain constructs what you see rather than recording it like a camera?
19. Why are prediction errors actually USEFUL even though they feel bad? Give two specific examples.
20. Design an experiment you could run at school to demonstrate the spotlight effect to your classmates. What would you have them do?